

EDUCATIONAL COGNITION ATLAS

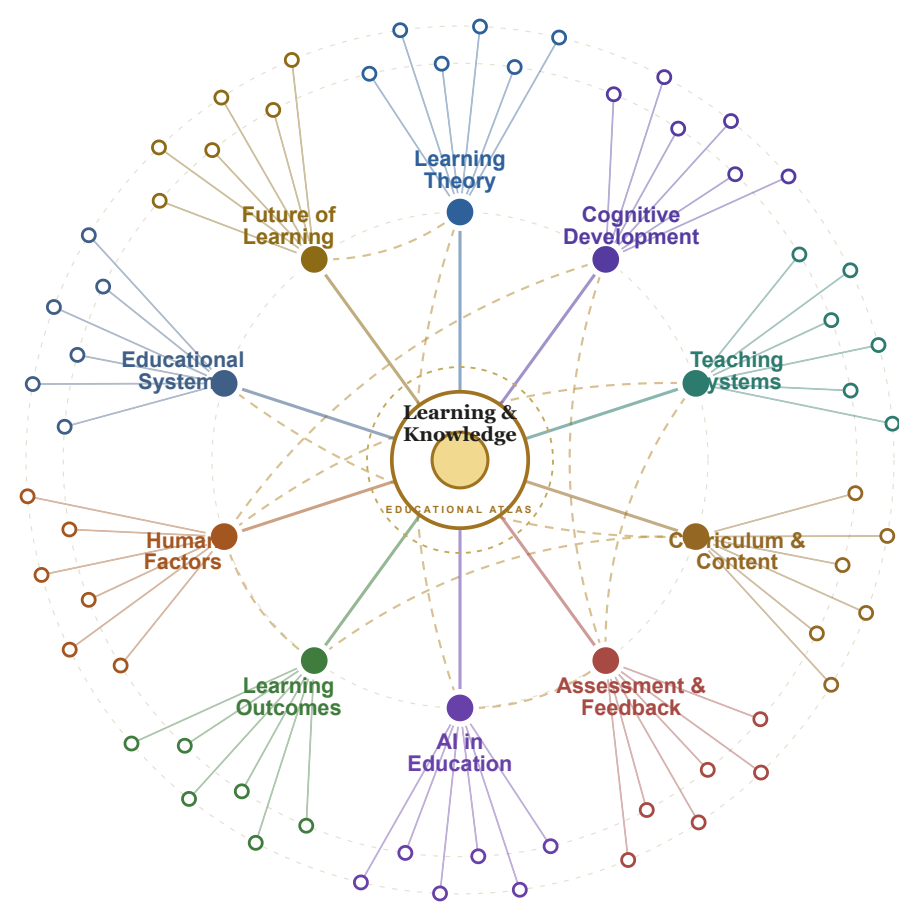
Learning & Knowledge Universe

A printed snapshot of how teaching, thinking, AI & human learning interconnect — ten educational domains, their branches, and the hidden adjacencies between them.

EDITION	PAGES	TRUNKS	BRANCHES	CROSS-LINKS
Print · A4	7	10	60	10
● Learning Theory	How learning happens		● Cognitive Development	How the mind builds
● Teaching Systems	How knowledge is delivered		● Curriculum & Content	What is learned
● Assessment & Feedback	How learning is measured		● AI in Education	Machine-assisted learning
● Learning Outcomes	What learning produces		● Human Factors	The learner inside
● Educational Systems	Structures around learning		● Future of Learning	Where learning is going

INTERACTIVE VERSION

<https://aurits.github.io/slide-desk/mindmap.html>



01 Learning Theory HOW LEARNING HAPPENS

- Constructivism** — Active Learning · Discovery · Schemas · Piaget · Vygotsky
- Behaviorism** — Reinforcement · Stimulus / Response · Drill & Practice · Skinner
- Cognitivism** — Information Processing · Schema Theory · Working Memory · Encoding
- Connectivism** — Networked Knowledge · Digital Learning · Distributed Cognition · Pattern Recognition
- Experiential** — Reflection · Action Learning · Practice · Kolb Cycle
- Social Learning** — Modelling · Imitation · Collaboration · Bandura

02 Cognitive Development HOW THE MIND BUILDS

- Memory** — Working Memory · Long-term Memory · Encoding · Retrieval · Knowledge Retention
- Attention** — Focus · Selective Attention · Task Switching · Sustained Effort
- Reasoning** — Deductive · Inductive · Analogical · Critical Thinking · Structured Logic
- Metacognition** — Self-Awareness · Reflection · Planning · Monitoring · Thinking about Thinking
- Problem Solving** — Heuristics · Analysis · Synthesis · Application
- Creativity** — Divergent Thinking · Convergent Thinking · Imagination · Generalization · Association

03 Teaching Systems HOW KNOWLEDGE IS DELIVERED

- Pedagogy** — Methods · Scaffolding · Differentiation · Inquiry-based
- Teacher Development** — Training · Mentorship · Professional Practice · Growth
- Classroom Management** — Engagement · Climate · Pacing · Discipline · Attendance
- Modes of Delivery** — Lectures · Discussion · Demonstration · Workshops · Teaching
- Resources** — Textbooks · Materials · Courses · Digital Tools
- Teacher ↔ Student** — Trust · Authority · Care · Strictness · Relationships

04 Curriculum & Content WHAT IS LEARNED

- Curriculum Design** — Standards · Sequencing · Outcomes Mapping · Goals
- Subjects** — STEM · Humanities · Arts · Languages
- Adaptive Curriculum** — Personalization · Branching Paths · Mastery Learning · Pacing
- Skill Pathways** — Foundational · Specialized · Cross-curricular · Skill Development
- Content Resources** — Textbooks · Open Resources · Digital Libraries · Research Papers
- Preparation** — Study Habits · Practice · Goal Setting · Continuity

05 Assessment & Feedback HOW LEARNING IS MEASURED

- Formative** — Quizzes · Check-ins · Observation · Discussion
- Summative** — Exams · Tests · Final Projects · Standardized
- Records** — Grades · Passes · Retakes · Attendance · Transcripts
- Feedback** — Reflection · Coaching · Adjustment · Iteration
- Criteria** — Output Quality · Process Visibility · Understanding Depth · Comparison
- Hidden Gaps** — Surface vs Deep · Knowledge Gaps · Misconceptions · Visibility of Reasoning

06 AI in Education MACHINE-ASSISTED LEARNING

AI Tutoring — Personalized Tutors · 24/7 Q&A · Conversational Practice · Worked Examples

Adaptive Learning — Pathways · Mastery Pacing · Difficulty Tuning · Personalization

AI Assessment — Auto-grading · Visibility of Reasoning · Surfacing Hidden Gaps · Quality Assurance

Prompting as Reasoning — Prompt Engineering · Prompt Recording · Workflow Standardization · AI Synthesis

AI Categorization — Structure Generation · AI-assisted Organization · Visual Synthesis · Mind Mapping

Ethics & Validation — Human Validation · Transparency · Accountability · Ethical AI Use · Equity

07 Learning Outcomes WHAT LEARNING PRODUCES

Knowledge — Recall · Understanding · Wisdom · Experience

Skills — Cognitive Skills · Practical Skills · Social Skills · Skill Development

Career — Employment · Salary · Specialization · Projects

Society — Citizenship · Responsibility · Contribution · Value

Lifelong Learning — Continuity · Reskilling · Future of Work · Adaptation

Meaning — Happiness · Self-Realization · Confidence · Purpose

08 Human Factors THE LEARNER INSIDE

Motivation — Curiosity · Goals · Drive · Effort

Emotion — Feeling · Confidence · Pressure · Relaxation · Happiness

Psychological Safety — Trust · Openness · Vulnerability · Belonging

Relationships — Teacher ↔ Student · Parent ↔ Child · Peers · Mentors

Abilities & Practice — Talent · Effort · Practice · Developing Abilities

Discipline — Strictness · Laziness · Self-Regulation · Procrastination

09 Educational Systems STRUCTURES AROUND LEARNING

Schools — Primary · Secondary · Charter · Private

Higher Education — Colleges · Universities · Research Labs · Postgraduate

Policy — Government · Ministry of Education · Rules & Regulations · Standards

Educational Leadership — Principals · Deans · Administrators · Decision Making

Organizational Learning — Continuity · Scaling · Operational Structure · Team Capability

Economics — Cost · Funding · Value Exchange · Productivity · Efficiency

10 Future of Learning WHERE LEARNING IS GOING

Educational Technology — Platforms · Digital Tools · Online Learning · Physical Learning · Digital / Physical

Research — Educational Research · Sample Data · Forecasting · Evidence-based

Meta-Cognition — Structure Discovery · Categorization Effects · Prompting as Reasoning · Self-Reflection

Emergent Insight — Unexpected Connections · Disproportionate Value · Hidden Variables · Color Coding

Equity & Access — Inclusion · Differentiation · Affordability · Reach

Lifelong Pathways — Adaptive Lifelong · Reskilling · Career Pivot · Self-directed

HIDDEN CROSS-LINKS — DASHED GOLD CURVES ON THE MAP

Learning Theory ↔ AI in Education

Teaching Systems ↔ Assessment & Feedback

Human Factors ↔ Learning Outcomes

Educational Systems ↔ Curriculum & Content

Cognitive Development ↔ Human Factors

Cognitive Development ↔ Assessment & Feedback

Curriculum & Content ↔ Learning Outcomes

AI in Education ↔ Assessment & Feedback

Future of Learning ↔ Learning Theory

Teaching Systems ↔ Human Factors

Prompt 01 THE INITIAL BRIEF

<https://aurits.github.io/slide-desk/mindmap.html>**DISCOVERY** *"Build a visual, interactive mindmap — not an essay — that reveals how concepts connect."*

You are an advanced systems-thinking, educational cognition, organizational design, and AI-structure synthesis assistant.

Your ONLY objective is to generate a highly visual, structured, interactive MINDMAP as a SINGLE self-contained HTML file.

DO NOT generate:

- essays
- long report sections
- academic-style explanations
- large textual summaries
- documentation-heavy outputs

The PRIMARY OUTPUT must be:

- a VISUAL MINDMAP
- highly organized
- interactive
- expandable/collapsible
- visually connected
- aesthetically clean
- easy to navigate
- focused on relationships and structures

The final deliverable must be:

- ONE standalone HTML file
- with embedded CSS and JavaScript
- no external dependencies required unless absolutely necessary
- immediately runnable in browser

CORE OBJECTIVE

The purpose of this exercise is NOT business implementation.

The purpose is:

- structure discovery
- concept synthesis
- relationship mapping
- trunk/branch identification
- AI-assisted categorization
- visualization of thinking patterns
- standardization of reasoning workflows

The mindmap should visually represent:

- how concepts connect
- which nodes are central ("trunks")
- which nodes are branches/leaves
- emotional vs logical structures
- AI vs human reasoning
- educational structures
- operational structures
- relationship density
- hidden conceptual intersections

IMPORTANT SESSION INSIGHTS

The following ideas are CRITICAL and should influence the structure:

- AI can improve outputs while hiding understanding gaps
- Education requires visibility of misunderstanding
- Work environments prioritize output quality
- Mindmaps reveal thinking structures
- Trunk concepts are more important than isolated leaves
- Emotional concepts are important structural signals
- Prompting itself represents human reasoning
- Categorization changes interpretation
- Valuable discoveries emerge from unexpected connections
- Human reasoning is nonlinear and associative
- AI should assist organization, not replace interpretation
- Structure discovery itself may become a leadership skill
- Some concepts carry disproportionately high value
- Relationship visualization is more important than summaries

VISUAL STYLE REQUIREMENTS

The visual style should be:

- Light background (NOT dark mode)
- Elegant, Minimalistic, Creative
- Professional, Spacious, Modern
- Easy to read

Use:

- soft color palettes
- thin connectors
- smooth animations
- rounded nodes
- layered hierarchy
- floating node structures
- subtle shadows
- curved relationship lines
- expandable clusters
- hover interactions

Typography:

- use modern creative fonts
- readable and elegant
- avoid overly technical styling

Add:

- small artistic illustrations/icons where appropriate

- subtle educational/AI/organizational symbolic visuals
- lightweight conceptual illustrations
- minimal visual clutter

The visual design should feel like:

- a creative strategic thinking board
- an AI-assisted cognition map
- a modern systems-thinking visualization
- a research-lab conceptual framework

MANDATORY STRUCTURE

The HTML mindmap MUST contain:

1. Central Root Node
2. Primary Trunk Categories
3. Secondary Branch Structures
4. Leaf Concepts
5. Relationship Connectors
6. Emotional/Abstract Layer
7. AI & Human Reasoning Layer
8. Organizational Structure Layer
9. Educational Structure Layer
10. Hidden Variable Connections

The structure should visually distinguish:

- trunk / branch / leaf nodes
- emotional / operational / AI / educational nodes

INTERACTION REQUIREMENTS

The HTML must support:

- zooming, panning
- node expansion/collapse
- hover highlighting
- smooth transitions
- relationship tracing
- node grouping
- clickable category focus

Optional:

- draggable nodes
- animated connectors
- mini-map overview

IMPORTANT OUTPUT CONSTRAINT

DO NOT turn the output into a written report.

The page should be:

~80-90% VISUAL MINDMAP
~10-20% supporting contextual labels

Keep textual explanations VERY SHORT.

BOTTOM EXPANDABLE SECTION

At the BOTTOM of the HTML page:

Create an expandable/collapsible section titled:

"AI Prompting & Reasoning Workflow"

Inside this section include:

1. The FULL prompt used
2. Prompt engineering logic
3. Why certain categories were created
4. How prompting influences structure
5. Differences between AI organization and human reasoning
6. Important limitations of AI-generated categorization
7. Human validation importance

This section should:

- remain collapsed by default
- expand smoothly
- visually separate from the main mindmap

COLOR STRUCTURE

Use simple soft colors:

Education -> Soft Blue
Human emotions/thinking -> Soft Orange
AI/Digital -> Soft Purple
Organization/Business -> Soft Green
Evaluation/Measurement -> Soft Red
Abstract/Philosophical -> Soft Gold

Trunk nodes:

- slightly darker
- slightly larger
- more visually emphasized

CONCEPTS TO INCLUDE

EDUCATION:

Knowledge, Learning, Understanding, Wisdom, Research, Experience, Teaching, Application, Evaluation, Exams, Tests, Curriculum, Lectures, Courses, Textbooks, Institutions, Schools, Colleges, Universities, Teachers, Students, Parents, Government, Ministry of Education, Rules & Regulations, Grades, Passes, Retakes, Attendance, Preparation, Goals, Process, Value, Career,

Society, Salary, Business, Projects, Future, Responsibility, Curiosity, Reflection, Emotion, Happiness, Continuity HUMAN THINKING & LEARNING: Trunk, Branch, Structure, Connection, Categorization, Color coding, Association, Visualization, Self-reflection, Comparison, Developing abilities, Thinking ability, Talent, Effort, Practice, Confidence, Creativity, Generalization, Knowledge gaps, Psychological safety, Pressure, Relaxation, Laziness, Strictness, Feeling, Relationships, Teacher/Student relationship, Parent/Child relationship AI & DIGITAL: AI, Digital tools, Digital/Physical, Online learning, Physical learning, AI-assisted organization, Prompt engineering, Workflow standardization, Prompt recording, AI categorization, AI synthesis, AI structure generation, AI-assisted evaluation, Visibility of reasoning, Hidden understanding gaps, Human validation, Quality assurance, Ethical AI usage, Transparency, Accountability ORGANIZATIONAL: Company, Business, Projects, Cost, Cost cutting, Value exchange, Growth, Scaling, Hiring ability, Evaluation, Management, Leadership, Decision making, Operational structure, Continuity, Organizational learning, Team capability, Risk, Productivity, Efficiency, Sample data, Forecasting, Systematic thinking,	Structured logic ----- FINAL GOAL Generate a deeply interconnected, visually elegant HTML mindmap that transforms these concepts into a structured intelligence system showing: - human thinking flows - conceptual relationships - educational structures - AI reasoning structures - organizational cognition - emotional influence patterns - trunk vs branch dynamics - visibility vs hidden structures The result should feel like: - a living conceptual ecosystem - a strategic cognition interface - a next-generation AI-assisted mindmap - a visual reasoning architecture - a leadership thinking framework FINAL OUTPUT: ONLY return the complete HTML file. DO NOT add explanations outside the HTML.
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REFINEMENT *"Reverse the layout — education dominates; all other categories become supporting branches."*

You over-produced explanatory sections and under-produced the actual EDUCATION-FOCUSED MINDMAP.

Revise the entire output.

The PRIMARY OBJECTIVE is NOT general conceptual discussion.

The PRIMARY OBJECTIVE is:

- a LARGE-SCALE EDUCATION-CENTERED VISUAL MINDMAP
- focused on EDUCATION SYSTEMS, LEARNING, HUMAN THINKING, AI IN EDUCATION, KNOWLEDGE DEVELOPMENT, REASONING STRUCTURES
- with the other categories only acting as SUPPORTING CONNECTION LAYERS
- visually dominant educational hierarchy
- extremely readable, easy to navigate
- zoomable, scrollable, spacious, visually clean

The current result still feels too much like:

- documentation
- essay sections
- conceptual article formatting
- textual reasoning panels

Reduce that dramatically.

The output should feel like:

- a strategic education cognition map
- an AI-assisted educational intelligence system
- a next-generation learning architecture visualization
- a visual educational ecosystem
- a structured knowledge universe

CRITICAL REVISION REQUIREMENTS

- 1. EDUCATION MUST DOMINATE THE MAP

The central structure should revolve around:

Education, Learning, Knowledge, Understanding, Thinking, Curriculum, Teaching, Students, Teachers, Educational Systems, Educational Evaluation, Educational Growth, Human Learning Processes, Cognitive Development, Knowledge Visibility, Learning Outcomes, AI in Education.

The other categories — AI, Organization, Evaluation, Human Emotion, Abstract Thinking — should mainly SUPPORT and CONNECT to educational structures.

Education should occupy the largest visual territory.

- 2. REDUCE WORDY PANELS

The lower explanation section is too large and too text-heavy. Instead:

- keep ONLY a SMALL expandable prompt section at the bottom
 - make it compact, scrollable, elegant
 - not dominant, max height constrained
 - visually secondary
- The mindmap itself must dominate the page.

- 3. MAKE THE MINDMAP MUCH CLEARER

Improve: node spacing, visual hierarchy, readability, label visibility, connector clarity, navigation clarity. Increase: node sizes, font sizes, spacing between major trunks, readability at zoomed-out levels. Avoid clutter.

- 4. MAKE EDUCATIONAL HIERARCHIES CLEAR

The visual hierarchy should clearly separate:

ROOT

- > Educational Domains
- > Learning Structures
- > Teaching Systems
- > Cognitive Processes
- > Evaluation Structures
- > AI-Assisted Learning
- > Outcomes
- > Human Factors

Use clear trunk -> branch -> leaf progression.

- 5. IMPROVE NAVIGATION

The map should feel smooth and intuitive.

MANDATORY: drag to move, wheel zoom, touch zoom, minimap, focus mode, smooth expand/collapse, hover highlights, readable connectors, centered navigation, intelligent spacing.

- 6. MAKE IT FEEL LIKE A LIVING EDUCATIONAL SYSTEM

The map should visually communicate growth, learning flow, knowledge movement, reasoning pathways, educational relationships, educational ecosystems, conceptual development. NOT just random nodes.

- 7. VISUAL STYLE

Keep: light background, soft academic aesthetic, elegant typography, modern educational visual language. Use: subtle artistic educational illustrations, soft symbolic icons, minimal cognitive visuals, elegant gradients, clean academic design. Avoid: dark mode, cyberpunk, overly technical UI, excessive

dashboards, harsh colors, dense panels.

- 8. FONT & NODE IMPROVEMENTS

Increase: readability, font scaling, node width, node padding, spacing consistency.

Major educational trunks should feel important, central, visually weighted.

- 9. STRUCTURAL INTELLIGENCE

The map should reveal:

- relationships between learning and AI
- relationships between teaching and evaluation
- relationships between thinking and understanding
- hidden educational gaps
- knowledge visibility
- educational feedback loops
- curriculum systems
- emotional impacts on learning
- organizational influence on education
- AI-assisted reasoning

- 10. HTML OUTPUT REQUIREMENTS

Return:

- ONE COMPLETE HTML FILE ONLY
- embedded CSS, embedded JavaScript
- self-contained, immediately runnable

NO markdown explanations.

NO essay sections.

NO external commentary.

- 11. IMPORTANT DESIGN CORRECTION

The current output still behaves like a webpage with a mindmap attached. Reverse this.

It should behave like:

A MASSIVE INTERACTIVE EDUCATIONAL MINDMAP

with only a small supporting prompt section.

- 12. MAKE THE PROMPT SECTION BETTER

At the bottom:

- create a compact expandable/collapsible section
- smooth animations
- elegant academic styling
- internally scrollable

Include:

- the original prompt used
 - AI reasoning notes
 - structural observations
 - educational insights
 - prompting workflow notes
- Keep it visually lightweight, not dominant, hidden by default.

- 13. EDUCATIONAL EXPANSION

Expand educational branches deeply. Examples:

Education

- > Learning Theory
- > Cognitive Development
- > Knowledge Retention
- > Classroom Systems
- > AI Tutoring
- > Curriculum Design
- > Human Motivation
- > Assessment Systems
- > Research
- > Educational Technology
- > Future of Learning
- > Adaptive Learning
- > Skill Development
- > Reflection
- > Educational Equity
- > Teacher Development
- > Knowledge Transfer
- > Creativity

- > Problem Solving

- > Educational Leadership

- > Lifelong Learning

Expand intelligently beyond the original words.

- 14. FINAL EXPERIENCE TARGET

The final result should feel like:

- an educational intelligence network
- a visual cognition engine
- an AI-assisted educational architecture
- a strategic learning ecosystem
- a conceptual educational atlas
- a future-of-learning interface

The experience should immediately communicate:

THIS IS ABOUT EDUCATION, LEARNING, THINKING, AND KNOWLEDGE STRUCTURES.

NOT generic conceptual categorization.